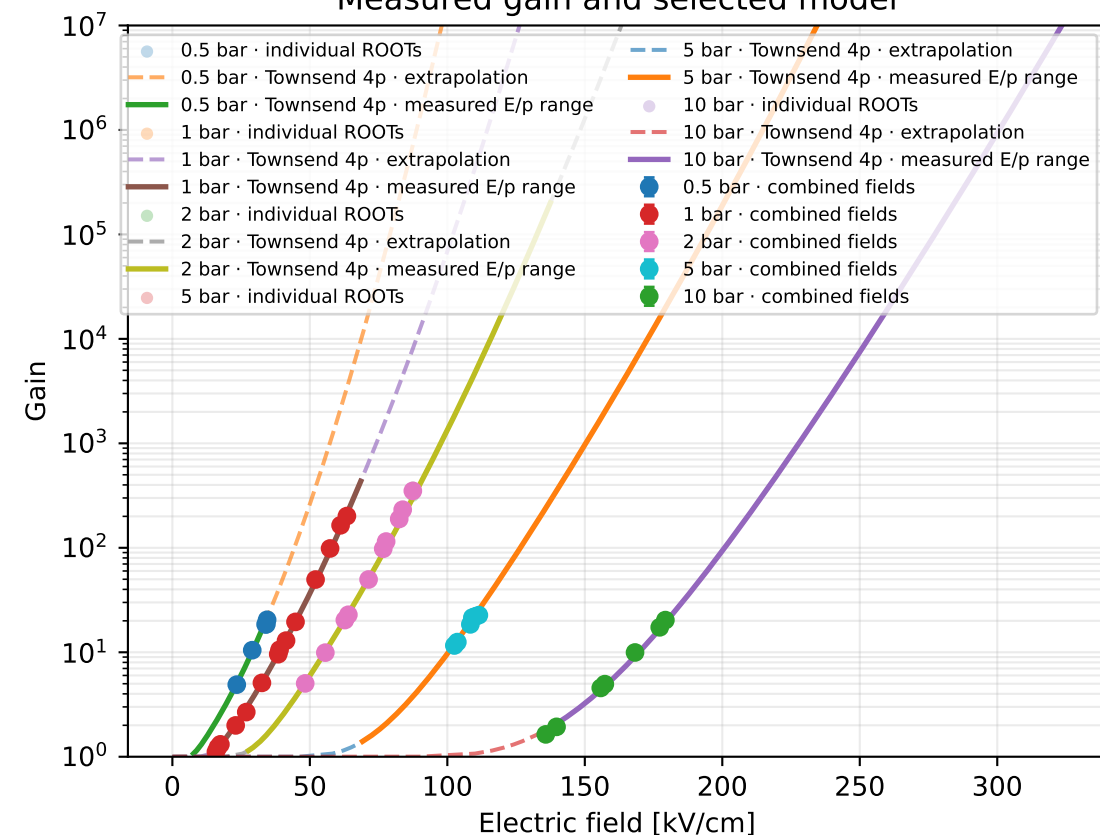
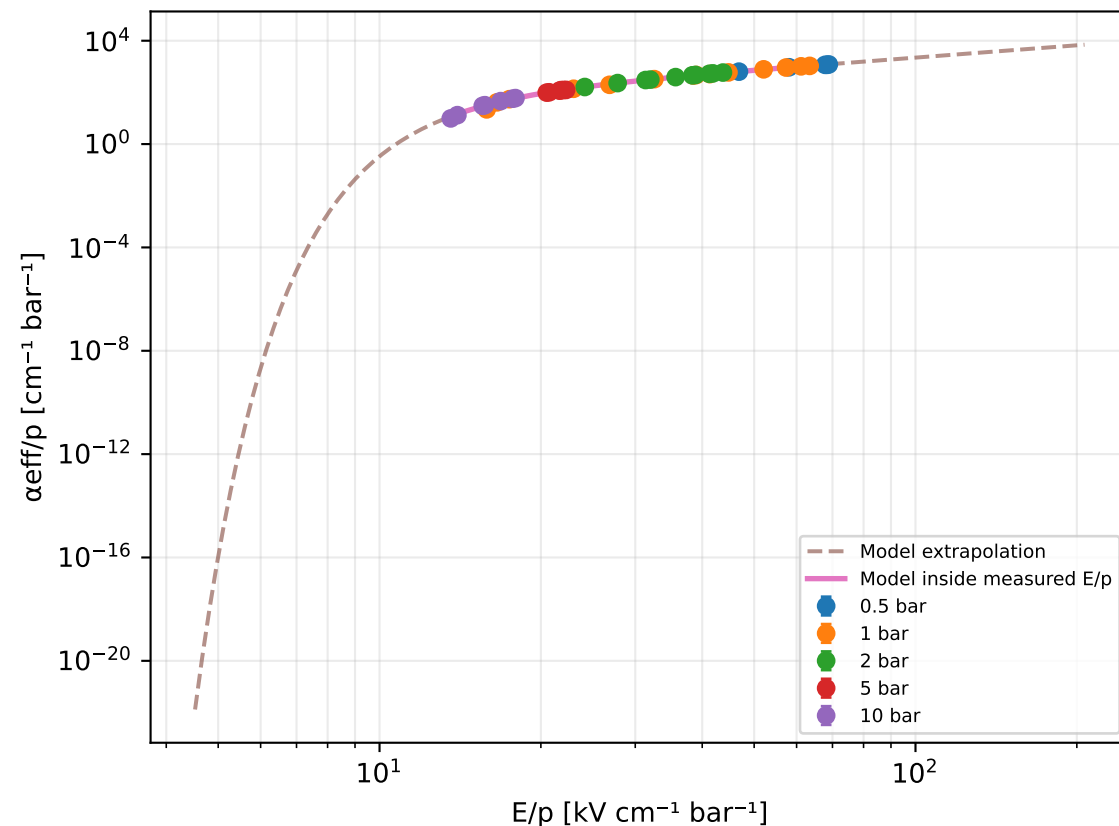


ArCF4 · Ar 95 % / CF4 5 % · gap 0.05 mm

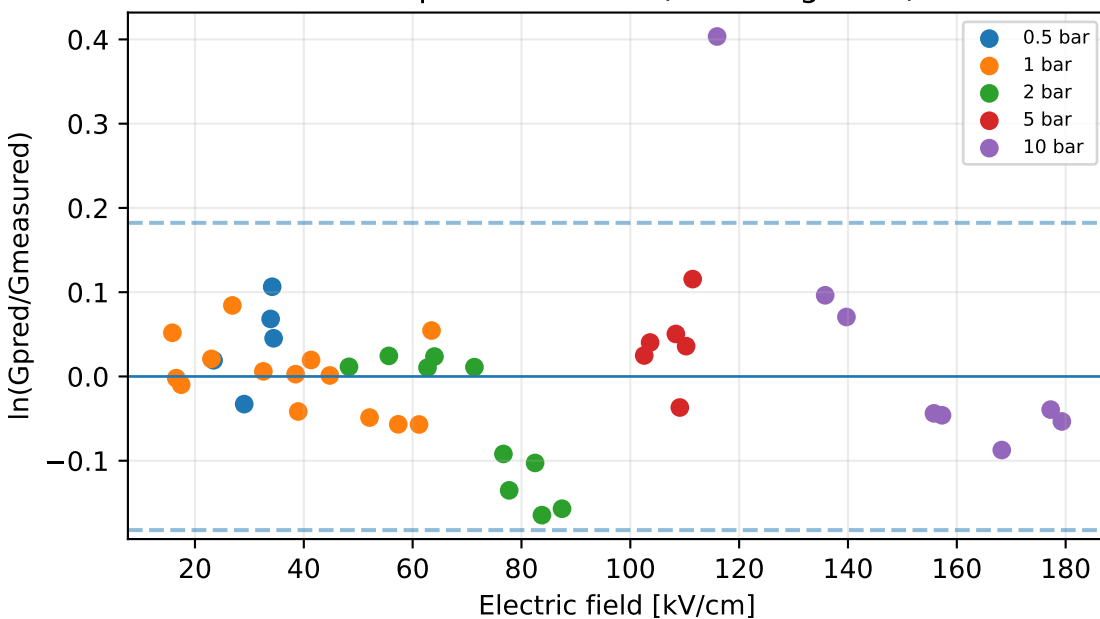
Measured gain and selected model



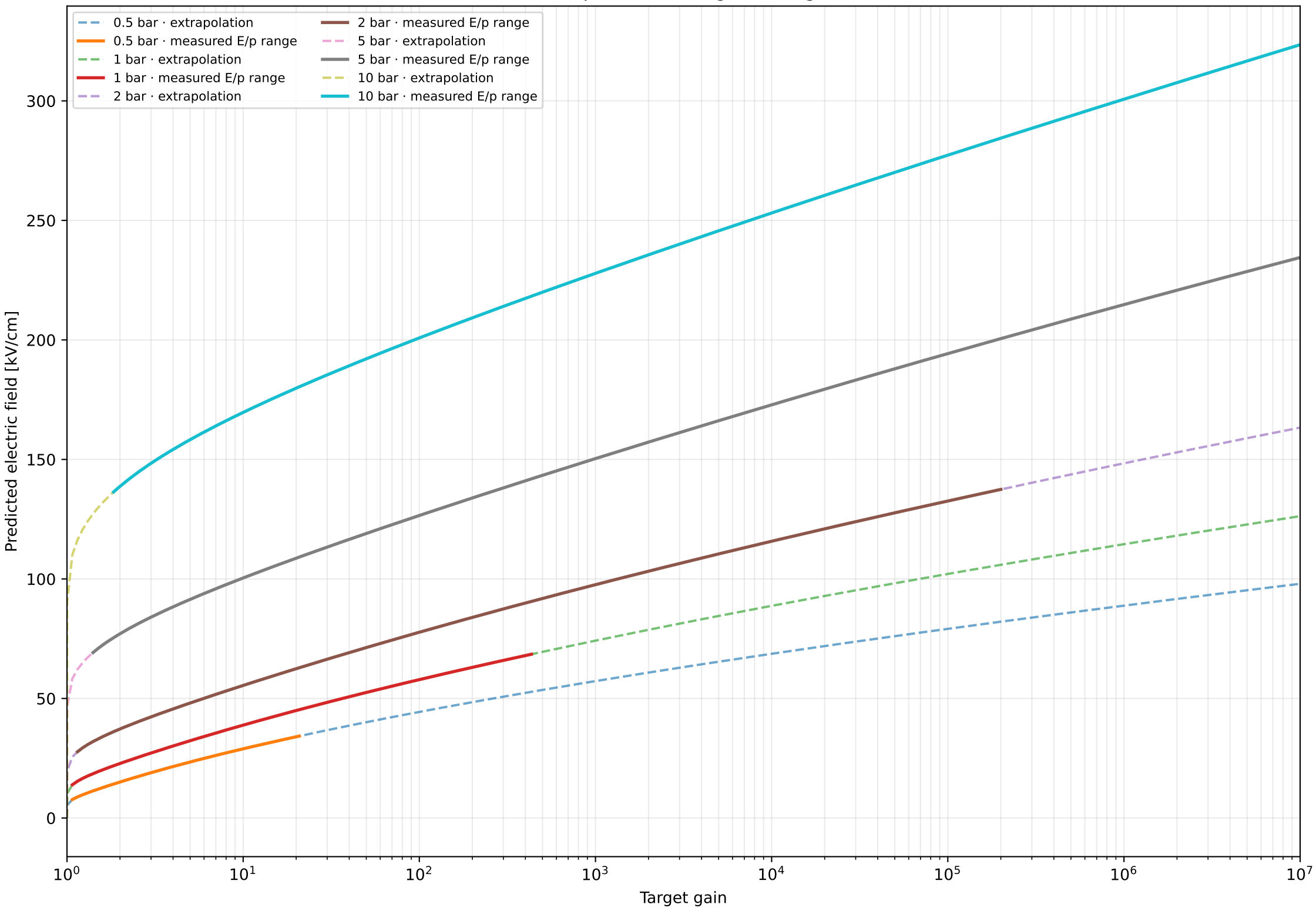
Reduced Townsend fit



Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 160
 Unique physical points: 43
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 4p
 Status: accepted
 Training RMSE: $\times 1.074$
 Cross-validation median: $\times 1.066$
 Cross-validation worst: $\times 1.183$
 Reduced-field range: 13.58–68.91 kV cm $^{-1}$ bar $^{-1}$
 A=1.6069, B=17.506, m=1.5722, n=2.9388
 Covariance condition: 3.73e+04

Field extrapolation from gain 1 to gain 10^7 

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
0.5	23.4186	46.8373	4.899554	0.147	317.829	448	1
0.5	29.0448	58.0895	10.46953	0.314	469.694	443	1
0.5	33.9053	67.8106	18.48787	0.554	583.423	371	1
0.5	34.1899	68.3799	18.53699	0.556	583.954	419	1
0.5	34.4529	68.9057	20.46809	0.613	603.773	376	1
1	15.8475	15.8475	1.116505	0.0334	22.0406	206	1
1	16.5634	16.5634	1.229391	0.0368	41.3037	279	1
1	17.4658	17.4658	1.314199	0.0394	54.6455	331	1
1	23.0201	23.0201	1.995215	0.0597	138.15	418	1
1	26.8704	26.8704	2.675768	0.0802	196.847	586	1
1	32.5658	32.5658	5.0937	0.153	325.601	619	1
1	38.493	38.493	9.583893	0.287	452.017	596	1
1	38.9793	38.9793	10.56714	0.317	471.55	566	1
1	41.3391	41.3391	12.92935	0.387	511.9	552	1
1	44.7778	44.7778	19.54032	0.586	594.496	620	1
1	52.0777	52.0777	49.6953	1.49	781.182	489	1
1	57.3526	57.3526	98.58667	3.53	918.187	300	1
1	61.173	61.173	164.27	6.12	1020.3	300	1
1	63.4709	63.4709	201.35	8.14	1061.01	300	1
2	48.3039	24.1519	5.034711	0.151	323.271	605	1
2	55.6291	27.8145	9.920103	0.297	458.913	776	1
2	62.7284	31.3642	20.33803	0.61	602.498	639	1
2	64.0084	32.0042	22.86233	0.685	625.898	661	1
2	71.355	35.6775	49.75688	1.49	781.43	654	1
2	76.6841	38.342	97.95563	2.94	916.903	586	1
2	77.7386	38.8693	114.8116	3.44	948.659	584	1
2	82.5014	41.2507	188.82	7.75	1048.16	300	1
2	83.7618	41.8809	231.64	9.6	1089.04	300	1
2	87.4515	43.7258	350.6955	4.31	1171.98	3300	11
5	102.553	20.5105	11.61499	0.09	490.459	14779	15

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
5	103.638	20.7277	12.51472	0.375	505.381	917	1
5	108.373	21.6746	18.49949	0.555	583.549	983	1
5	109.099	21.8197	21.48959	0.644	613.514	913	1
5	110.253	22.0507	22.08497	0.662	618.979	918	1
5	111.464	22.2927	22.66824	0.679	624.193	847	1
10	115.953	11.5953	0.7623252	0.00236	—	180991	94
10	135.807	13.5807	1.637029	0.0491	98.5766	1912	1
10	139.717	13.9717	1.932469	0.058	131.76	1851	1
10	155.8	15.58	4.557692	0.137	303.363	1612	1
10	157.278	15.7278	4.952168	0.149	319.965	1568	1
10	168.26	16.826	9.954414	0.298	459.603	1382	1
10	177.248	17.7248	17.33744	0.52	570.574	1218	1
10	179.301	17.9301	20.32256	0.61	602.346	1330	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-173	×1.16	×1.48	yes	—
Townsend 3p	3	-157	×1.08	×1.54	yes	—
Townsend 4p	4	-213	×1.07	×1.18	yes	—